

① Input of D Flip Flop is
 $D = X \oplus Q$.

Since $Q_{n+1} = D$.

$$\therefore Q_{n+1} = X \oplus Q_n.$$

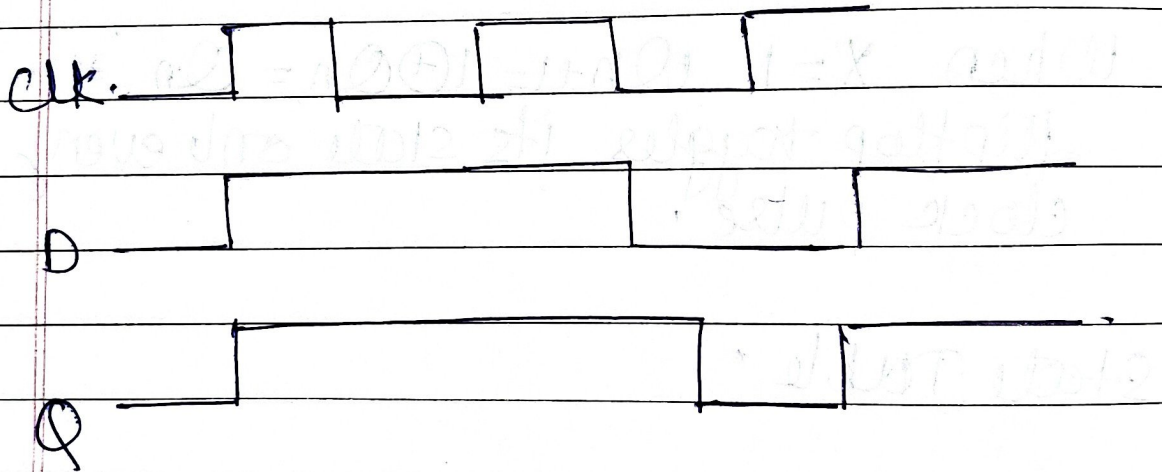
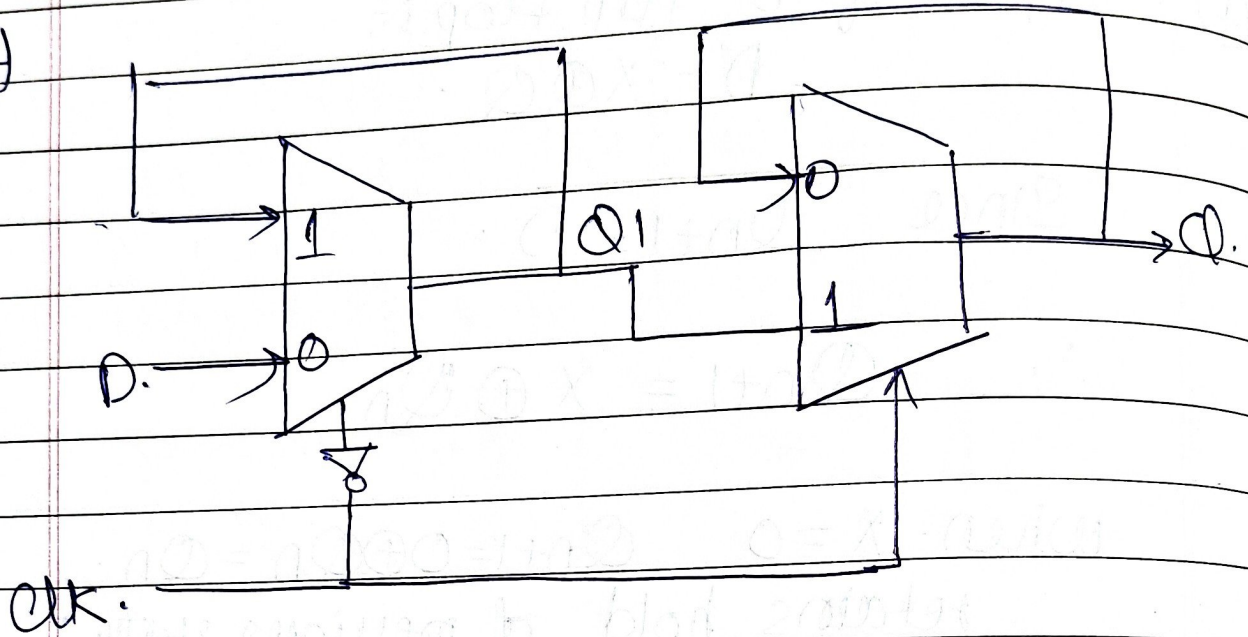
When $X = 0$ $Q_{n+1} = 0 \oplus Q_n = Q_n$.
retains hold of previous state.

When $X = 1$ $Q_{n+1} = 1 \oplus Q_n = \overline{Q_n}$.
flipflop toggles its state on every clock pulse.

State Table.

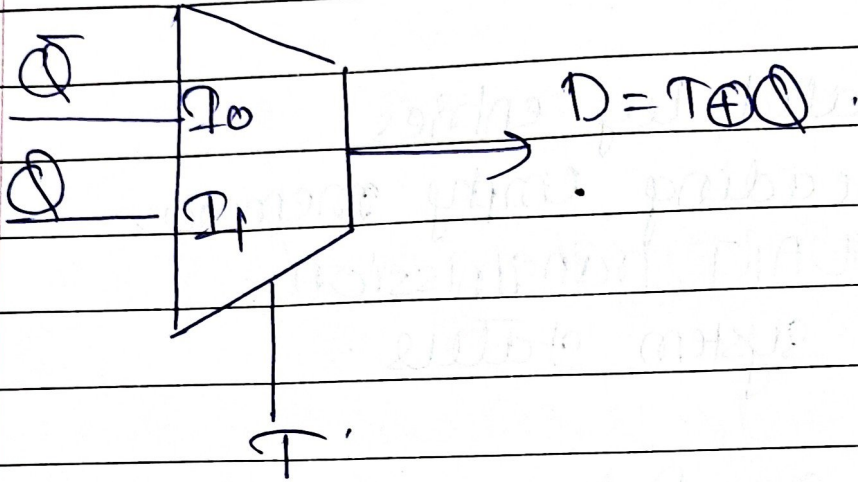
X	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

(2)

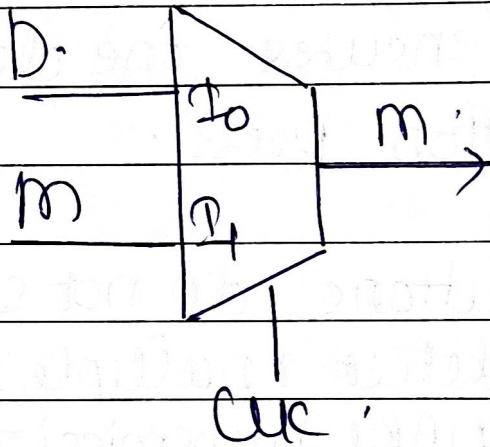


1	1	1	1
0	0	0	0
1	1	1	1
0	0	0	0
1	1	1	1
0	0	0	0

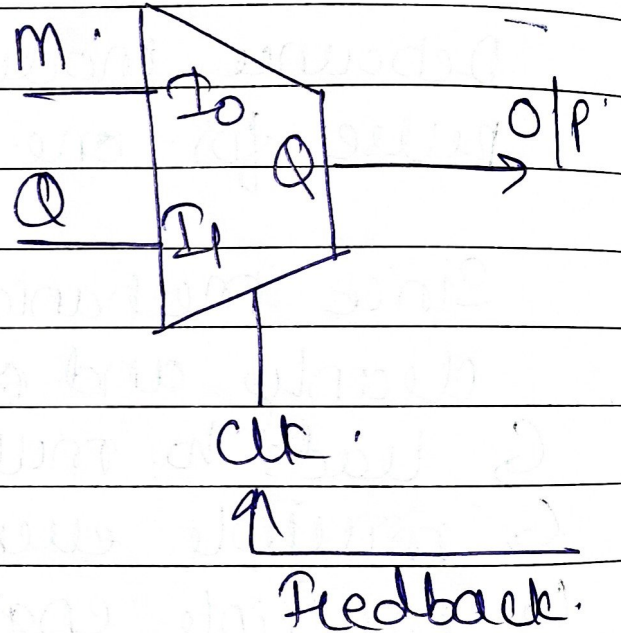
③



MASTER LATCH



SLAVE LATCH



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ASSIGNMENT 4.

PAGE NO.

DATE

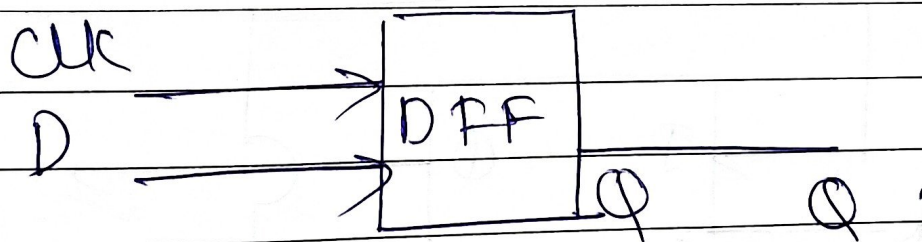
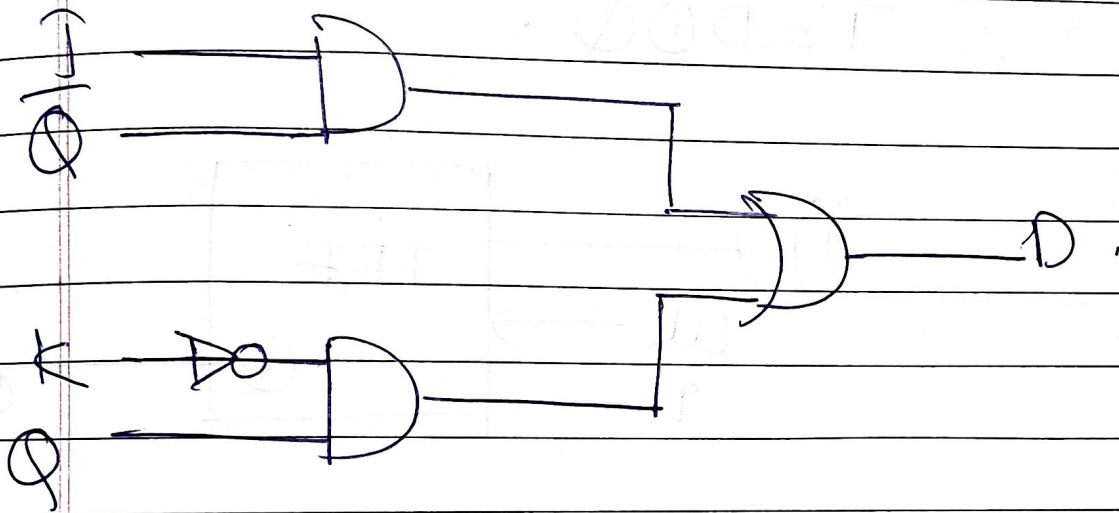
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(1) a. D FF to a JK FF.

D Flip Flop : $Q_{n+1} = D$.

JK Flip Flop : $Q_{n+1} = J\bar{Q} + KQ$.

$$D = J\bar{Q} + KQ.$$



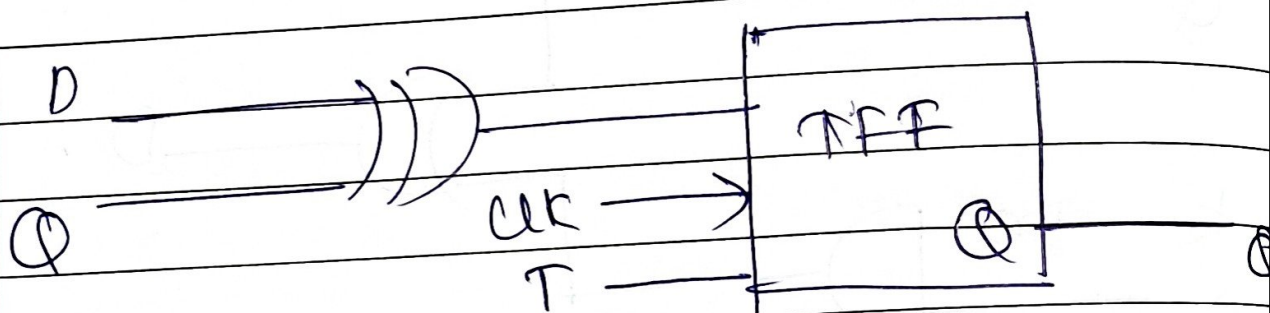
4(b) T FlipFlop to a D FlipFlop.

$$T \text{ FF} : Q_{n+1} = T \oplus Q$$

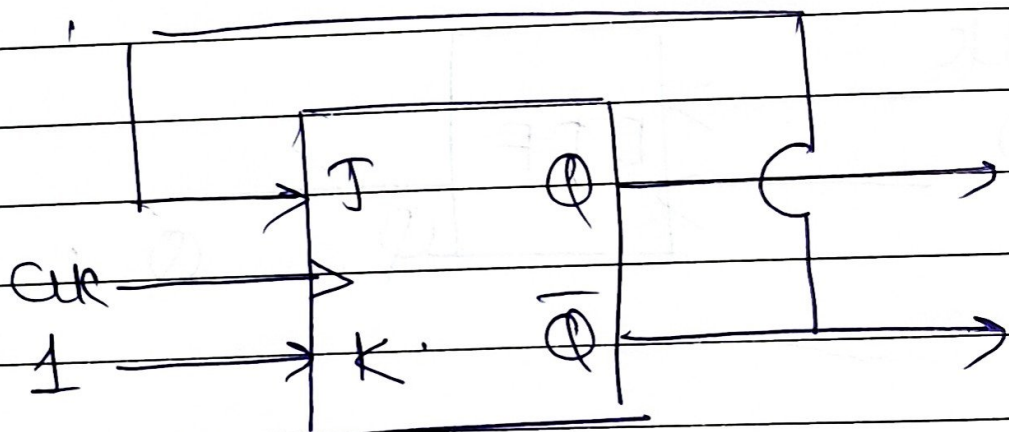
$$D \text{ FF} : Q_{n+1} = D$$

$$D = T \oplus Q$$

$$\therefore T = D \oplus Q$$



(5)



JK FlipFlop Characteristic Table.

J	K	Next State.
0	0	No change
0	1	Reset (0)
1	0	Set (1)
1	1	Toggle.

Since $J = Q'$.

Present Q	$J = Q'$	K	Next Q.
0	1	1	1 (Toggle)
1	0	1	0 (Reset)

Sequence for 6 Clock Pulses.

$$Q_0 = 0.$$

Clock Pulse	Q	$J = Q'$	K	Next Q
Initial	0	1	1	1
1	1	0	1	0
2	0	1	1	1
3	1	0	1	0
4	0	1	1	1
5	1	0	1	0
6	0	1	1	1

Output Sequence:

Starting from the initial created state:

$$Q = 1, 0, 1, 0, 1, 0$$

$$0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0.$$